

KASBench Full Disclosure Report

Preamble

This report documents the results obtained from an execution of the KASBench Kubernetes Autoscaling Benchmark.

Talk about completely random design.

Talk about run duration and number of trials

Insert a section for each autoscaler (how configured)

Benchmark Metadata

Benchmark Name	KASBench
Version	v1.0
Testing Dates	2026-09-18 20:33:12 UTC to 2026-09-19 10:16:10 UTC
Reporting Date	2026-09-22 13:52:03 UTC
Benchmark Application	GlobeCo
Cloud Provider	Amazon Web Services
Region	us-east-1
Benchmark Runner	c6a.large (amd64)
Control Plane	m8i.large (amd64)
Worker Nodes AMD	1 x c6a.4xlarge
Worker Nodes ARM	1 x c6g.4xlarge
AMD Worker AMI ID	ami-0244585558b2aebd7
ARM Worker AMI ID	ami-052bbac83b5bboab9
AMD Runner AMI ID	ami-03a891c9de365d954
Kubernetes Version	1.36.1
Autoscalers Evaluated	Kubernetes Horizontal Pod Autoscaler (HPA), Kubernetes Event-Driven Autoscaler (KEDA), Kubernetes Vertical Pod Autoscaler (VPA), No autoscaler
Trials Per Autoscaler	4

Software artifacts

Benchmark Components

Component	Image_original	ImageID
KASBench Controller	kasbench/ kasbench-controller:0.0.6	sha256:0816294976161de3279756584c5ee4061eafb860d80e436ca0a33dc99cedabc4
KASBench Runner	kasbench/ kasbench-runner:0.2.22	sha256:a28c2f8b2a9ca961e2186dob5e078e373065427b406bc13752a98f9b84d63085
KASBench Load Generator	kasbench/ kasbench-load-generator:0.0.21	sha256:5e6ad9d312140a7f6e19f5792aa68b4a1c12d1e2c6999e98f0bd5f24549bca4d

Helm Charts

Name	Version	URL
aws-ebs-csi-driver	2.66.0	https://kubernetes-sigs.github.io/aws-ebs-csi-driver
aws-efs-csi-driver	4.5.0	https://kubernetes-sigs.github.io/aws-efs-csi-driver/
globeco	0.1.51	https://kasbench.github.io/globeco-helm
keda	2.20.2	https://kedacore.github.io/charts
prometheus	29.31.1	https://prometheus-community.github.io/helm-charts

Stateful Services

Stateful Set	Image
postgres	postgres:18.0 docker.io/library/postgres sha256:41fc5342eeffa6cc2ccda736aaf034bbbb7c3df0fdb81516eba1ba33f360162c
apache/kafka	apache/kafka:4.0.0 docker.io/apache/kafka sha256:3f7b939115cd4872e9cee9369d80bd69712fde55f9902f46d793f64848dedc75
mongo	mongo:6.0 docker.io/library/mongo sha256:8b6d8f5bbedb25cb73517b65cf99f13aeb75ad5b157a56c479287a840bbad3ac
redis	redis:7-alpine docker.io/library/redis sha256:520775a41a63e77e06c73e35d2fd9cc15921a609516818796b4ecbb813078bc7

Stateless Services

Deployment	Image
globeco-allocation-service	<code>kasbench/globeco-allocation-service-server:1.0.3</code> <code>docker.io/kasbench/globeco-allocation-service-server</code> <code>sha256:0e6750db5d9e6227554a346f12d0556ea4c62a6047786f2395a539c74218cd5e</code>
globeco-confirmation-service	<code>kasbench/globeco-confirmation-service:1.0.6</code> <code>docker.io/kasbench/globeco-confirmation-service</code> <code>sha256:5d3267895a231b9210451f253boed783301520f036c4eab085b736405e3d7d64</code>
globeco-debug-tools	<code>kasbench/globeco-debug-tools:latest</code> <code>docker.io/kasbench/globeco-debug-tools</code> <code>sha256:9f279ca1ccaf4d118c3de3587bb7946198ac6a14569e3990ea978fdc4f644032</code>
globeco-execution-service	<code>kasbench/globeco-execution-service:1.0.7</code> <code>docker.io/kasbench/globeco-execution-service</code> <code>sha256:07eb1ec41a44e1b6a80a50b17c4121372bd95a6c51879b08afbe79a28533c9e6</code>
globeco-fix-engine	<code>kasbench/globeco-fix-engine:1.0.13</code> <code>docker.io/kasbench/globeco-fix-engine</code> <code>sha256:1705dd116250a607d73e1fab8bb6f44f67ea289fa848d4a841d64fe924c1e5d7</code>
globeco-order-generation-service	<code>kasbench/globeco-order-generation-service:1.0.6</code> <code>docker.io/kasbench/globeco-order-generation-service</code> <code>sha256:a7541d05c1e720573394cb41b443067a05bbb7db75e69e240df1808c5fed5f14</code>
globeco-order-service	<code>kasbench/globeco-order-service:1.0.4</code> <code>docker.io/kasbench/globeco-order-service</code> <code>sha256:1600b3c42ace32ddf0296468173bc835344e4c885e0b528f4d4e71e51f457b7f</code>
globeco-portfolio-accounting-service	<code>kasbench/globeco-portfolio-accounting-service-server:1.0.2</code> <code>docker.io/kasbench/globeco-portfolio-accounting-service-server</code> <code>sha256:2b5646e0ccd9a5c4bf4bbce2749f6325c668802175cfc551c36c24fa36dee69d</code>
globeco-portfolio-management-portal	<code>kasbench/globeco-portfolio-management-portal:1.0.5</code> <code>docker.io/kasbench/globeco-portfolio-management-portal</code> <code>sha256:a65a6022cb244c4bafoeac2ad1f8ddd626ec4ecf0481947e639a0924feb1193</code>
globeco-portfolio-service	<code>kasbench/globeco-portfolio-service:1.0.4</code> <code>docker.io/kasbench/globeco-portfolio-service</code> <code>sha256:b0561599453606420eb4e904f31214c9c62c5a580c5ca95b25e59f67eca16dd7</code>
	<code>kasbench/globeco-portfolio-service-high-cpu:1.0.4</code> <code>docker.io/kasbench/globeco-portfolio-service-high-cpu</code> <code>sha256:1a7821e35baf1814f2a7fddbd446ca59b2deaa86fe0a822a5400a27fb59be49c</code>
globeco-pricing-service	<code>kasbench/globeco-pricing-service:1.0.2</code> <code>docker.io/kasbench/globeco-pricing-service</code> <code>sha256:2509c2a2e69712fa9a559b21a6f820ee537300592e6b77794c7135f282c06dc9</code>
globeco-security-service	<code>kasbench/globeco-security-service:1.0.1</code> <code>docker.io/kasbench/globeco-security-service</code> <code>sha256:7066be0ea6f76bd60d1ba3c24a30e748c9c767a5eb13a2859204ab765e632241</code>
globeco-trade-service	<code>kasbench/globeco-trade-service:1.0.4</code> <code>docker.io/kasbench/globeco-trade-service</code> <code>sha256:9c8f610f5fc34dac3eefc396a0f38f9ce4e24c22dea2b1a7acb7726962b26cb4</code>

Raw Performance Metrics

Summarized Performance Metrics by Trial

Trial ID	Autoscaler	Requests	Mean Response Time (ms)	Failures	Failure Rate	Round-trip Completion (%)
trial0001	HPA	271092	100	21	0.008%	1.4%
trial0002	HPA	269018	96	10	0.004%	1.1%
trial0003	VPA	270496	79	11	0.004%	2.2%
trial0004	KEDA	263695	73	168	0.064%	6.2%
trial0005	NONE	259293	205	55	0.021%	1.7%
trial0006	VPA	268839	82	1929	0.718%	2.5%
trial0007	NONE	263339	165	195	0.074%	1.0%
trial0008	HPA	269791	96	21	0.008%	1.1%
trial0009	KEDA	271437	79	4	0.001%	5.9%
trial0010	VPA	269663	75	5	0.002%	2.5%
trial0011	VPA	264706	77	734	0.277%	2.3%
trial0012	NONE	192960	188	16	0.008%	1.8%
trial0013	KEDA	263246	102	130	0.049%	6.8%
trial0014	HPA	270773	84	8	0.003%	1.2%
trial0015	KEDA	270186	94	112	0.041%	6.2%
trial0016	NONE	214979	240	165	0.077%	2.5%

Summarized Performance Metrics by Autoscaler

Autoscaler	Mean Response Time (ms)	Failure Rate	Round-trip Completion (%)
HPA	94	0.006%	1.2%
KEDA	87	0.039%	6.3%
NONE	200	0.045%	1.7%
VPA	78	0.250%	2.4%

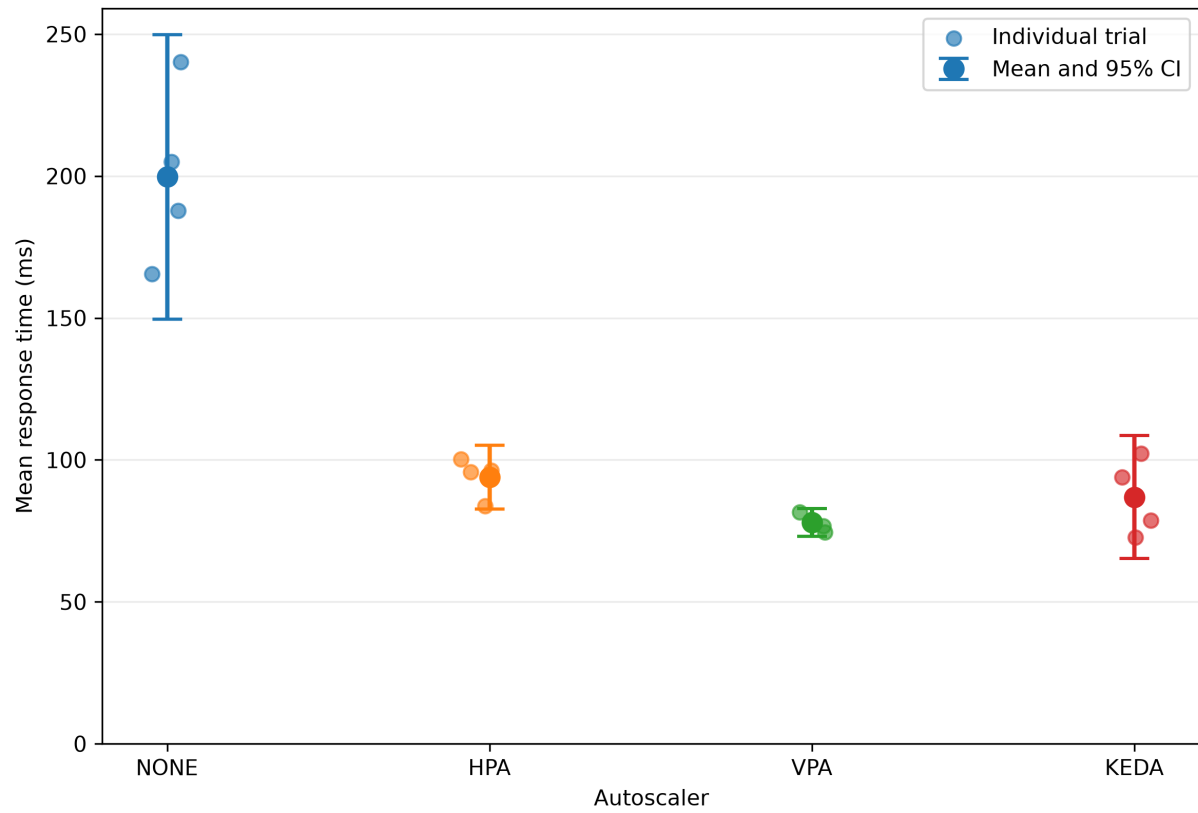


Figure 1. Mean response time by autoscaler.

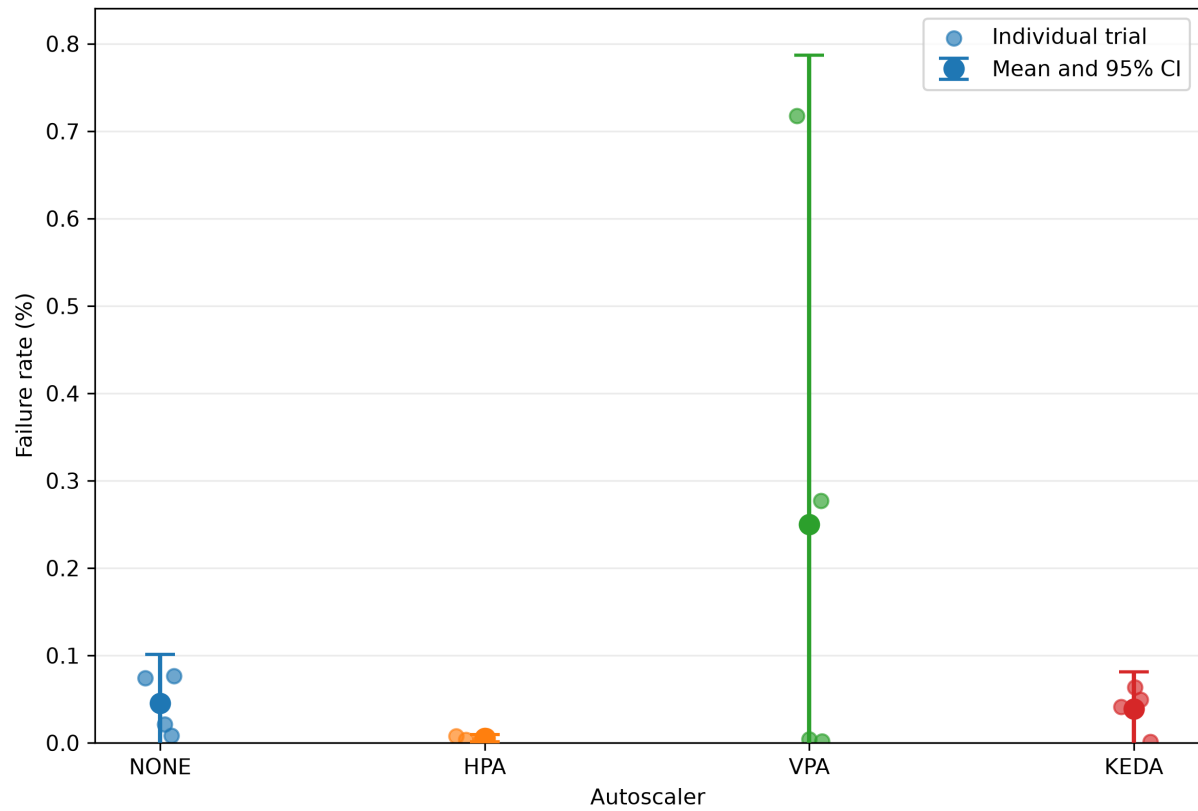


Figure 2. Mean failure rate by autoscaler.

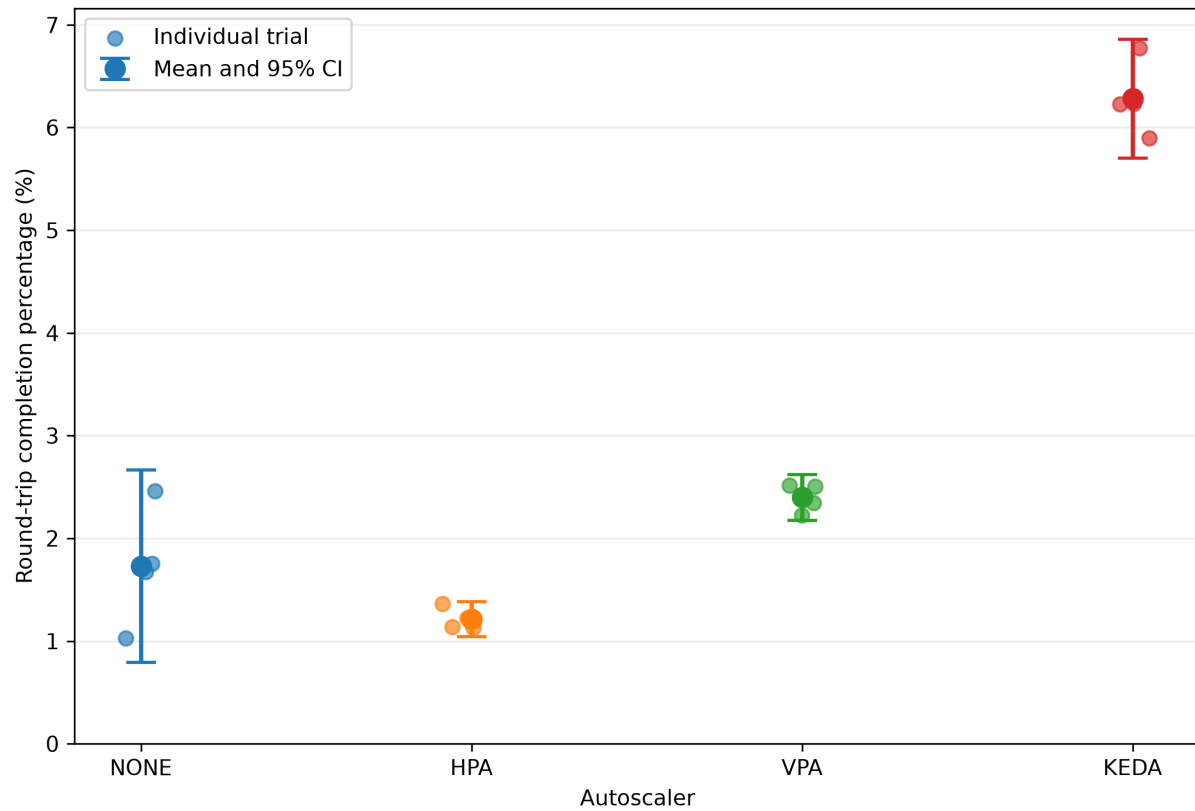


Figure 3. Mean round trip percentage by autoscaler.